

Notes: Edmans (JFE, 2011)
Does the Stock Market Fully Value Intangibles? Employee Satisfaction and Equity Prices

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1 Big picture

One-sentence summary. This paper studies whether employee satisfaction is associated with superior long-run stock returns and whether the stock market fully prices this form of intangible human capital. Using the list of the “100 Best Companies to Work For in America,” Edmans shows that firms with high employee satisfaction earn significantly positive abnormal returns over 1984–2009, with performance that is robust to industry, factor, characteristic, and outlier controls.

Core research question. The paper asks whether employee satisfaction is merely a costly perk for workers, or whether it is a productive intangible asset that improves shareholder value but is only slowly incorporated into prices.

Main finding. A value-weighted portfolio of the *Best Companies* earns a monthly four-factor alpha of about 0.29% over the risk-free rate, equivalent to roughly 3.5% per year, and still earns about 0.17% per month after industry adjustment. Equal-weighted returns are similar or stronger.

Economic interpretation. Employee satisfaction appears to be a value-relevant intangible asset. However, the market does not fully capitalize it immediately, even though the list is public, highly visible after 1998, and focuses on large firms. The return predictability therefore looks like long-run underreaction to human-capital-related intangibles.

2 Incremental contribution

2.1 Contribution to human-capital theories of the firm

Traditional theories of the firm treat employees largely as costs to be minimized. In that view, high employee satisfaction may indicate overpayment, underwork, weak monitoring, or managerial slack. More recent human-capital-centered theories argue that employees are organizational assets whose motivation, retention, creativity, and client relationships can create firm value.

The paper contributes by providing direct stock-market evidence that high employee satisfaction is positively associated with shareholder returns. This supports the modern view of the firm in which human capital is central to value creation.

2.2 Contribution to the market efficiency and intangibles literature

Prior studies show that the market underprices R&D, advertising, patent citations, software development, and other intangible assets. A common explanation is that investors lack salient information about intangible quality. Edmans uses a highly public measure - the *Best Companies* list - to test whether lack of information is the main explanation.

The key incremental contribution is that mispricing remains even when the intangible is visible, independently certified, and associated with large firms. This implies that non-incorporation of intangibles is not only a disclosure problem; it may also arise because investors use valuation frameworks that are poorly suited to intangible assets.

2.3 Contribution to socially responsible investing

The paper also contributes to the socially responsible investing literature. Classical portfolio theory implies that an SRI screen should reduce expected returns by restricting the investor’s opportunity set. Edmans shows that at least one SRI screen - employee satisfaction - improves performance. The reason is that the screen may help investors identify underpriced intangible assets rather than merely exclude investments.

3 Data and institutional setting

3.1 The Best Companies list

The main data source is the list of the “100 Best Companies to Work For in America.” The list was first published in book form in 1984, updated in 1993, and published annually in *Fortune* from 1998 onward. The list is based primarily on employee survey responses and an independent assessment of firm culture, benefits, credibility, respect, fairness, pride, and camaraderie.

A key advantage of this measure is that it captures an output of employee relations - actual satisfaction - rather than only observable inputs such as benefits or diversity programs. This makes it more informative than simple policy-based screens.

3.2 Portfolio construction

The paper forms a portfolio after the list becomes public, thereby excluding the immediate event-study response to list announcement. The 1984 portfolio starts on April 1, 1984; the 1993 portfolio starts on March 1, 1993; and from 1998 onward each portfolio starts on February 1, after the January *Fortune* release.

The core portfolio, Portfolio I, is updated whenever a new list is available. It includes publicly traded firms on the most recent list. The paper studies both equal-weighted and value-weighted returns.

3.3 Sample characteristics

The firms in the list are typically large. For example, the mean market capitalization is about \$4 billion for the 1984 list and about \$21 billion for the 1998 list. This is important because many anomalies are concentrated in small firms. The evidence here is therefore more economically meaningful and more plausibly tradable.

4 Empirical design

4.1 Four-factor portfolio regressions

The main return specification is a Carhart four-factor regression:

$$R_t = \alpha + \beta_{MKT}MKT_t + \beta_{HML}HML_t + \beta_{SMB}SMB_t + \beta_{MOM}MOM_t + \varepsilon_t. \quad (1)$$

Here R_t is the return on the *Best Companies* portfolio in month t in excess of one of three benchmarks: the risk-free rate, an industry-matched benchmark, or a characteristics-matched benchmark. The intercept α measures abnormal performance after controlling for factor exposures.

4.2 Benchmark choices

The paper uses three benchmarks:

- **Risk-free benchmark:** tests whether the portfolio earns a standard four-factor alpha.
- **Industry benchmark:** subtracts returns of industry-matched portfolios to ensure results are not driven by industries that happened to perform well.
- **Characteristics benchmark:** subtracts returns of Daniel, Grinblatt, Titman, and Wermers-style portfolios matched on size, book-to-market, and momentum.

Using all three benchmarks is conservative because the characteristics benchmark already adjusts for some patterns that the four-factor model also captures.

4.3 Fama-MacBeth characteristics regressions

To control for a broader set of firm characteristics, the paper estimates monthly cross-sectional regressions:

$$R_{it} = a_0 + a_1 BC_{it} + a_2 Z_{it} + e_{it}, \quad (2)$$

where BC_{it} is a dummy for whether the firm is included in the most recent *Best Companies* list, and Z_{it} includes size, book-to-market, dividend yield, past returns, trading volume, and price.

The coefficient a_1 measures whether inclusion in the list predicts future returns after controlling for standard return predictors.

5 Main return results

5.1 Full sample results

Over 1984–2009, the *Best Companies* portfolio earns economically and statistically significant abnormal returns. The value-weighted portfolio earns an alpha of 0.29% per month over the risk-free rate and 0.17% per month over industry benchmarks. Equal-weighted returns are similarly strong, with a 0.31% monthly alpha over the risk-free rate and 0.20% over industry benchmarks.

Interpretation. These results suggest that employee satisfaction is not merely a redistributive transfer from shareholders to workers. Instead, the stock market appears to underreact to information about satisfied employees.

5.2 Post-1998 results

From 1998 onward, the list is published annually in *Fortune*. This subsample is useful because the list is highly visible. The paper finds that abnormal returns remain positive and in some

specifications are even larger. This is inconsistent with the view that prior non-incorporation of intangibles arises only because investors lack access to information.

5.3 Robustness

The paper shows that the results are robust to:

- value-weighting and equal-weighting;
- industry-adjusted and characteristics-adjusted benchmarks;
- winsorizing extreme returns;
- controls for firm characteristics in Fama-MacBeth regressions;
- excluding the immediate announcement-window response.

The winsorization tests are especially important because they show that the alpha is not driven by a small number of extreme stocks.

6 Mechanism: earnings surprises and market underreaction

6.1 Earnings surprise tests

The paper tests whether satisfaction eventually becomes visible through tangible accounting performance. The earnings surprise regression is:

$$Surprise_{it} = b_0 + b_1 BC_{it} + b_2 Z_{it} + e_{it}. \quad (3)$$

The dependent variable is the one-year or two-year earnings surprise, or the long-term growth surprise. The key coefficient b_1 asks whether *Best Companies* firms subsequently deliver more positive earnings news than other firms.

6.2 Results and interpretation

The *Best Companies* firms exhibit more positive earnings surprises and more positive earnings announcement returns. Their three-day abnormal returns around earnings announcements are significantly higher than those of other firms. Since the return window around earnings announcements is short, this evidence is less sensitive to benchmark-model concerns.

Economic message. The evidence supports a mispricing channel: employee satisfaction is an intangible that the market initially underprices, but later partially recognizes when it is converted into tangible outcomes such as earnings news.

7 Longevity of outperformance

If the return predictability reflects mispricing, one expects abnormal returns to decline as the market learns or as the underlying satisfaction asset changes. The paper studies cumulative abnormal

returns (CARs), buy-and-hold abnormal returns (BHARs), firms that remain on the list for five years, and portfolios of firms dropped from the list.

The key result is that the abnormal returns persist for several years but largely disappear by the fifth year. This pattern is consistent with gradual learning and with the fact that employee satisfaction is persistent but not permanent.

8 Alternative explanations and causal interpretation

The paper discusses several alternative explanations:

- **SRI demand pressure:** SRI funds may buy listed firms and push up prices. The paper shows that such demand can explain only a tiny fraction of the alpha.
- **Initial undervaluation from negative beliefs:** The market may initially penalize employee-friendly firms. This is inconsistent with evidence that the *Best Companies* firms trade at valuation premiums rather than discounts.
- **Reverse causality:** Employees may be more satisfied because they know future performance will be strong. The paper argues this is unlikely because of the long lag between surveys and returns and evidence that employees generally do not forecast firm returns well.
- **Omitted variables:** Satisfaction may proxy for unobserved good management. This cannot be fully ruled out, but even under this interpretation the market still underprices publicly available information about valuable intangibles.

Important nuance. The study is not a randomized experiment, so the cleanest causal statement is not that increasing satisfaction mechanically raises stock returns. The most robust statement is that the *Best Companies* list identifies valuable, underpriced intangible assets - whether those assets are satisfaction itself, good management, or employee confidence.

9 Economic takeaway

The paper provides three broad lessons.

1. **Human capital matters.** Employee satisfaction is associated with superior future performance and is not necessarily managerial slack.
2. **Markets underprice intangibles.** Even highly visible information about intangible assets may not be fully incorporated into prices.
3. **Some SRI screens can add value.** Employee welfare can be a useful signal of future returns, not merely a constraint on the investment set.

10 Figures and tables

10.1 Table 1: Summary statistics and Table 2: Summary characteristics

Table 1

Summary statistics.

The second column details the number of the 100 Best Companies to Work For in America (Best Companies) that had returns available on CRSP for at least one month between publication of the list of that year, and the subsequent list. The third column gives the number of new public companies added to the Best Companies list of that year. The fourth column contains the number of companies on the previous Best Companies list which no longer feature in the current list or are no longer public. The sample period is 1984–2009.

Year of list	Best Companies	Added	Dropped
1984	78		
1993	69	30	39
1998	70	34	33
1999	68	26	28
2000	60	20	28
2001	55	15	20
2002	55	13	13
2003	61	14	8
2004	57	11	15
2005	58	11	10
2006	50	9	14

(a) Table 1: Summary statistics.

Table 2

Summary characteristics.

Summary characteristics for the 74 companies in the 1984 “100 Best Companies to Work For in America” (Levring, Moskowitz, and Katz, 1984) list that were public on April 1, 1984, and the 69 companies in the 1998 list published in *Fortune* that were public on February 1, 1998. The first two items are taken from CRSP at the end of March 1984 (January 1998, respectively). The last three items are based on CRSP and Compustat data for 1997 (1983), missing for companies that were not traded in 1997 (1983), and excluded for companies for which only the ADRs are traded.

	# obs	Mean	Median	Std. dev.	Min	Max
<i>1984 List</i>						
Market Cap (\$ bn)	74	3.99	1.25	9.48	0	69.47
Price (\$)	74	37.43	33.88	19.64	5.91	113.75
Dividend yield (%)	69	2.45	2.23	2.03	0	9.06
Market/book	69	2.41	1.95	1.82	0.68	10.80
Intangibles as a % of total assets (%)	69	0.91	0	2.15	0	10.35
<i>1998 List</i>						
Market Cap (\$ bn)	69	21.33	5.24	39.52	0.03	204.59
Price (\$)	69	51.35	44.22	25.47	5.38	127.56
Dividend yield (%)	63	1.60	1.03	4.31	0	34.26
Market/book	63	5.20	4.13	4.22	-5.34	20.91
Intangibles as a % of total assets (%)	63	5.23	0.08	7.75	0	29.97

(b) Table 2: Summary characteristics.

Extracted tables from Edmans (2011).

Read. Table 1 shows the number of public *Best Companies* firms in each list year, along with additions and deletions. Table 2 shows that list firms are large, high market-to-book companies with little recognized intangible capital on the balance sheet.

Economic message. The portfolio is not a small-cap anomaly. The firms are large and visible, which makes the subsequent return predictability more striking.

10.2 Table 3: Risk-adjusted returns and Table 4: Risk-adjusted returns from 1998

Table 3

Risk-adjusted returns.

Monthly regressions of returns to a portfolio of the “100 Best Companies to Work For in America” on the four Carhart (1997) factors, *MKT*, *HML*, *SMB*, and *MOM*. The dependent variable is the portfolio return less either the risk-free rate, the industry-matched portfolio return, or the characteristics-matched portfolio return. Panel A contains equal-weighted returns and Panel B contains value-weighted returns. The alpha is the excess risk-adjusted return. *t*-Statistics are in parentheses. The sample period is April 1984–December 2009.

	Excess returns over		
	Risk-free	Industry	Characteristics
<i>Panel A: Equal-weighted</i>			
α	0.31 (3.34)***	0.20 (2.76)***	0.24 (2.94)***
β_{MKT}	1.08 (41.01)***	0.06 (3.55)***	0.09 (3.69)***
β_{HML}	0.03 (0.70)	0.09 (3.22)***	0.01 (0.45)
β_{SMB}	0.17 (3.66)***	0.15 (5.70)***	0.05 (1.39)
β_{MOM}	−0.15 (−6.36)***	−0.07 (−3.39)***	−0.09 (−4.80)***
<i>Panel B: Value-weighted</i>			
α	0.29 (2.59)***	0.17 (2.28)**	0.15 (2.15)**
β_{MKT}	1.00 (35.68)***	−0.04 (−0.18)	0.01 (0.59)
β_{HML}	−0.37 (−7.64)***	−0.03 (−0.76)	−0.11 (−3.32)***
β_{SMB}	−0.17 (−3.64)***	−0.21 (−6.63)***	−0.03 (−0.88)
β_{MOM}	−0.06 (−1.78)*	−0.02 (−0.81)	−0.04 (−2.11)**
# obs	309	309	309

*: Significant at the 10% level; **: Significant at the 5% level; ***: Significant at the 1% level.

Table 4

Risk-adjusted returns from 1998.

Monthly regressions of returns to a portfolio of the “100 Best Companies to Work For in America” on the four Carhart (1997) factors, *MKT*, *HML*, *SMB*, and *MOM*. The dependent variable is the portfolio return less either the risk-free rate, the industry-matched portfolio return, or the characteristics-matched portfolio return. Panel A contains equal-weighted returns and Panel B contains value-weighted returns. The alpha is the excess risk-adjusted return. *t*-Statistics are in parentheses. The sample period is February 1998–December 2009.

	Excess returns over		
	Risk-free	Industry	Characteristics
<i>Panel A: Equal-weighted</i>			
α	0.44 (2.89)***	0.31 (2.62)***	0.43 (3.46)***
<i>Panel B: Value-weighted</i>			
α	0.32 (1.65)	0.19 (1.50)	0.16 (1.35)
# obs	143	143	143

***: Significant at the 1% level.

(a) Table 3: Risk-adjusted returns.

(b) Table 4: Risk-adjusted returns from 1998.

Main abnormal return results.

Read. Table 3 documents significant positive four-factor alphas over the full sample. Table 4 repeats the analysis after 1998, when the list was published in *Fortune* and thus became highly salient.

Economic message. The excess returns survive risk adjustment and remain visible after the list becomes public and prominent, challenging the view that the market fully prices salient intangibles.

10.3 Table 5: Risk-adjusted returns of winsorized portfolios and Table 6: Characteristics regressions

Table 5

Risk-adjusted returns of winsorized portfolios.

Monthly regressions of returns to a portfolio of the "100 Best Companies to Work For in America" on the four Carhart (1997) factors, *MBT*, *HML*, *SMB*, and *MOM*. The returns of the Best Companies are winsorized at the 5% and 100 - 5% levels across the sample period. The dependent variable is the winsorized portfolio return less either the risk-free rate, the industry-matched portfolio return, or the characteristics-matched portfolio return. Panel A contains equal-weighted returns and Panel B contains value-weighted returns. The alpha is the excess risk-adjusted return. *t*-Statistics are in parentheses. The sample period is April 1984–December 2009 for the left-hand column, and February 1998–December 2009 for the right-hand column.

	<i>x</i> =5			<i>x</i> =10		
	Risk-free	Industry	Characteristics	Risk-free	Industry	Characteristics
<i>Panel A: Equal-weighted</i>						
<i>α</i>	0.35 (3.45)***	0.23 (2.80)***	0.28 (3.10)***	0.40 (4.18)***	0.29 (3.36)***	0.33 (3.76)***
<i>Panel B: Value-weighted</i>						
<i>α</i>	0.35 (3.33)***	0.23 (3.16)***	0.20 (2.66)***	0.40 (3.93)***	0.28 (3.61)***	0.25 (3.15)***
# obs	309	309	309	309	309	309

***: Significant at the 1% level.

(a) Table 5: Risk-adjusted returns of winsorized portfolios.

Table 6

Characteristics regressions.

Monthly regressions of individual stock returns on a dummy variable for whether the firm was in the most recent list of the "100 Best Companies to Work For in America" (*BC*) and the characteristics used in Brennan, Chordia, and Subrahmanyam (1998). *SIZE* is the log of the firm's market capitalization (in billions) in month *t* - 2. *BM* is the log of the firm's book-to-market ratio as of the calendar year-end before the most recent June. *YIELD* is the firm's dividend yield as of the calendar year-end before the most recent June. *RET2-3*, *RET4-6* and *RET7-12* are the logs of the compounded returns in, respectively, month *t* - 3 to month *t* - 2, month *t* - 6 to month *t* - 4, and month *t* - 12 to month *t* - 7. *DVOL* is the dollar trading volume (in millions) in month *t* - 2. *PRC* is the price at the end of month *t* - 2. *t*-Statistics are in parentheses. The sample period is April 1984–December 2009 for the left-hand column, and February 1998–December 2009 for the right-hand column.

	1984–2009		1998–2009	
	Raw	Industry-adjusted	Raw	Industry-adjusted
<i>BC</i>	0.39 (3.68)***	0.34 (3.58)***	0.34 (1.87)*	0.27 (1.72)*
<i>SIZE</i>	0.14 (1.86)*	0.15 (2.43)**	0.04 (0.34)	0.05 (0.43)
<i>BM</i>	0.21 (4.41)***	0.22 (5.79)***	0.09 (1.12)	0.10 (1.61)
<i>YIELD</i>	-0.03 (-2.45)**	-0.03 (-3.24)***	0.00 (0.20)	-0.00 (-0.28)
<i>RET2-3</i>	0.77 (2.64)***	0.39 (1.40)	1.04 (1.92)*	0.53 (1.04)
<i>RET4-6</i>	0.73 (2.90)***	0.49 (2.07)**	0.88 (1.89)*	0.45 (1.03)
<i>RET7-12</i>	0.89 (5.19)***	0.67 (4.13)***	0.58 (1.93)*	0.31 (1.08)
<i>DVOL</i>	-0.10 (-1.51)	-0.11 (-1.90)*	-0.02 (-0.17)	-0.05 (-0.51)
<i>PRC</i>	-0.29 (-2.66)***	-0.23 (-2.20)**	-0.40 (-2.16)**	-0.25 (-1.35)
Constant	2.04 (6.08)***	1.27 (3.01)***	1.59 (2.84)***	1.39 (1.95)*
# obs	1,691,492	1,673,440	819,956	813,707
Number of groups	309	309	143	143

*: Significant at the 10% level; **: Significant at the 5% level;

***: Significant at the 1% level.

most a portion of the abnormal returns. Since profits are persistent and thus affect stock returns only to the extent they are unexpected, I follow Core, Guay, and Rusticus (2006) and Giroud and Mueller (2011) and study earnings surprises. Using similar methodology to these papers, I run the following regression:

$$\text{Surprise}_{it} = b_0 + b_1 X_{it} + b_2 Z_{it-j} + \varepsilon_{it} \quad (3)$$

Surprise is the one- or two-year earnings surprise, or the long-term growth surprise. The one-year earnings surprise is the actual earnings per share (EPS) for the fiscal year ending in year *t* minus the median Institutional Brokers' Estimate System (I/B/E/S) analyst forecast, deflated by the

(b) Table 6: Characteristics regressions.

Robustness to outliers and firm characteristics.

Read. Table 5 shows that winsorizing extreme returns does not eliminate the alpha. Table 6 shows that the *Best Companies* dummy remains positive after controlling for size, book-to-market, dividend yield, past returns, dollar volume, and price.

Economic message. The return predictability is not an artifact of outliers or standard observable characteristics.

10.4 Table 7: Earnings surprises and Table 8: Earnings announcement returns

Table 7

Earnings surprises.

Regressions of earnings surprises on a dummy variable for whether the firm was in the most recent list of the “100 Best Companies to Work For in America” (*BC*) and controls (*BM*, log book-to-market and *SIZE*, log market equity) calculated at the previous year-end. The 1- (2-) year earnings surprise is the actual EPS minus the I/B/E/S median analyst forecast 8 (20) months prior to the end of the forecast period, scaled by the stock price. The long-term growth surprise is the actual five-year annualized EPS growth rate minus the I/B/E/S median analyst long-term growth forecast from 56 months earlier. The Best Company dummy and control variables are taken from the same month as the I/B/E/S median forecast. Panel A (B) contains the results for 1- (2-) year earnings surprises; Panel C contains the results for long-term growth surprises. All coefficients are multiplied by 1,000. All regressions include year fixed effects and a constant, not reported for brevity. *t*-Statistics are in parentheses. The sample period is April 1984–December 2009.

	(1)	(2)	(3)
<i>Panel A: 1-Year earnings</i>			
<i>BC</i>	3.63 (5.26)***	3.17 (4.60)***	−1.14 (−1.63)
<i>BM</i>		−1.21 (−12.26)***	−0.41 (−4.01)***
<i>SIZE</i>			1.80 (31.26)***
# obs	75,813	72,164	72,164
<i>Panel B: 2-Year earnings</i>			
<i>BC</i>	3.89 (4.69)***	4.02 (4.84)***	−0.10 (−0.12)
<i>BM</i>		0.41 (3.00)***	1.23 (8.80)***
<i>SIZE</i>			1.93 (23.82)***
# obs	51,076	49,156	49,156
<i>Panel C: Long-term growth</i>			
<i>BC</i>	2.27 (4.08)***	3.55 (6.37)***	1.46 (2.57)***
<i>BM</i>		2.82	3.34

(a) Table 7: Earnings surprises.

Table 8

Earnings announcement returns.

(−1,+1) abnormal returns to quarterly earnings announcements. Abnormal returns are calculated above a market model in which the coefficients are estimated over a 255-day period ending 46 days before the earnings announcement. Panel A compares the average announcement returns to firms included in the most recent list of the “100 Best Companies to Work For in America” with the returns to all other firms. Panel B regresses announcement returns on a dummy variable for whether the firm was in the most recent Best Companies list (*BC*) and controls (*BM*, log book-to-market and *SIZE*, log market equity) calculated at the previous year-end. These regressions include year fixed effects and a constant, not reported for brevity. *t*-Statistics are in parentheses. The sample period is April 1984–December 2009.

<i>Panel A: Univariate comparisons</i>			
	Best Company		Other firms
CAR	0.36		0.08
# obs	5,241		311,328
t-Stat (difference from 0)	(40.57)***		(5.01)***
t-Stat (difference in means)	(2.20)**		
<i>Panel B: Regressions</i>			
	(1)	(2)	(3)
BC	0.29 (2.36)**	0.43 (3.49)***	0.36 (2.83)***
BM		0.31 (17.17)***	0.33 (17.37)***
SIZE			0.03 (3.12)***
# obs	316,569	296,826	296,826

: Significant at the 5% level; *: Significant at the 1% level.

one-third of firms drop off the list each year. If a firm's

(b) Table 8: Earnings announcement returns.

Evidence on tangible channels.

Read. Table 7 shows that *Best Companies* firms generate more positive earnings surprises, especially over longer horizons. Table 8 shows that *Best Companies* firms earn higher abnormal returns around earnings announcements.

Economic message. Satisfaction-related intangibles appear to become priced when they later materialize in earnings news.

10.5 Table 9: Longevity analysis and Table 10: List of employment funds

Table 9

Longevity analysis.

Panel A calculates cumulative abnormal returns (CAR) to portfolios of the "100 Best Companies to Work For in America" in calendar time. The abnormal return of stock in event month t is calculated by subtracting its benchmark return t months after list inclusion. The CAR through month t is an arithmetic sum of the abnormal returns from months 1 through t . Panel B calculates the buy-and-hold returns (BHAR). It first geometrically compounds the unadjusted returns of a Best Company from month 1–12, 13–24, etc. and then subtracts the geometrically compounded benchmark return over the same period. Panel C calculates the BHAR to companies that remain on the list for the following five years, winsorizing at the 5th and 95th percentiles. Panel D contains monthly regressions of the returns of Portfolios I and III on the four Calvert (1997) factors, AMT, JMT, SMB, and MOM. Portfolio II is the original 1984–1998 Best Companies list and is not updated with subsequent lists. Portfolio III contains only companies dropped from a prior list. The alpha is the excess risk-adjusted return. t -Statistics are in parentheses. The sample period is April 1984–December 2009 for Panels A–C and given in the headings for Panel D.

Panel A: CARs		CAR over				
Month	Market	Industry	Characteristics			
6	2.11	2.21	1.56			
12	5.57	4.72	4.76			
18	7.26	6.65	5.80			
24	10.36	8.66	8.82			
30	11.50	10.36	9.32			
36	15.73	12.88	12.54			
42	18.63	15.92	14.09			
48	21.03	17.33	16.03			
54	24.10	20.57	18.68			
60	24.21	20.68	18.66			
Panel B: BHARs		BHAR over				
Months	Market	Industry	Characteristics			
1–12	5.07 (2.11)**	4.53 (1.94)*	4.84 (1.90)*			
13–24	2.66 (1.88)*	2.24 (1.73)*	2.66 (1.91)*			
25–36	3.32 (2.41)**	2.63 (2.05)*	1.87 (1.36)			
37–48	3.35 (2.44)**	2.90 (2.25)**	1.94 (1.61)			
49–60	0.33 (0.24)	0.81 (0.62)	0.27 (0.21)			
Panel C: BHARs of firms on list for 5 years		BHAR over				
Months	Market	Industry	Characteristics			
1–12	9.47 (5.28)***	6.97 (4.34)***	6.81 (3.63)***			
13–24	6.63 (3.81)***	4.30 (2.82)***	5.41 (3.18)***			
25–36	3.10 (1.90)*	2.13 (1.40)	2.45 (1.50)			
37–48	2.95 (1.95)*	2.28 (1.62)	2.38 (1.68)*			
49–60	−0.36 (−0.23)	−0.07 (−0.04)	−0.82 (−0.56)			
Panel D: Alphas		1984–2009: excess returns over				
	Risk-free	Industry	Characteristics	1998–2009: excess returns over		
				Risk-free	Industry	Characteristics
Equal-weighted						
α, II	0.23 (2.15)**	0.12 (1.44)	0.13 (1.41)	0.44 (2.97)***	0.28 (2.55)**	0.34 (2.50)**
α, III	0.30 (2.68)***	0.20 (2.16)**	0.21 (2.36)**	0.34 (1.75)*	0.26 (1.82)*	0.27 (1.96)*
Value-weighted						
α, II	0.23 (2.31)**	0.16 (1.85)*	0.15 (2.36)**	0.24 (1.77)	0.08 (0.62)	0.11 (0.87)
α, III	0.16 (1.44)	0.04 (0.34)	0.04 (0.63)	−0.21 (−1.08)	−0.29 (−2.13)**	−0.19 (−1.44)
# obs	309 for II, 202 for III			143 for II, 131 for III		

*: Significant at the 10% level; **: Significant at the 5% level; ***: Significant at the 1% level.

Table 10

List of employment funds.

SRI funds that invest in domestic equity and use labor or employment screens. The main data sources are the Social Investment Forum and SocialFunds. Any conflicts were resolved by reading the fund prospectus or calling the fund. (P) denotes that the fund employs a Positive Investment screen on labor or employment, and (R) denotes a Restricted Investment screen.

AHA Socially Responsible Equity (P)	LKCM Acquisas Small Cap (R)
Applesed (P)	LKCM Acquisas Value (R)
Calvert Aggressive Allocation (P)	MMA Praxis Core Stock (P)
Calvert Capital Accumulation (P)	MMA Praxis Growth Index (P)
Calvert Conservative Allocation (P)	MMA Praxis Intermediate Income (P)
Calvert Global Alternative Energy (P)	MMA Praxis International (P)
Calvert International Opportunities (P)	MMA Praxis Small Cap (P)
Calvert Large Cap Growth (P)	MMA Praxis Value Index (P)
Calvert Mid Cap Value (P)	Neuberger Berman Socially Responsible (P)
Calvert Moderate Allocation (P)	New Alternatives (P)
Calvert New Vision Small Cap (P)	Parnassus (P)
Calvert Small Cap Value (P)	Parnassus Mid-Cap (P)
Calvert Social Index (P)	Parnassus Small-Cap (P)
Calvert Social Investment Balanced (P)	Parnassus Workplace (P)
Calvert Social Investment Enhanced Equity (P)	Pax World Balanced (P)
Calvert Social Investment Equity (P)	Pax World Growth (P)
Calvert World Values International (P)	Pax World High Yield (P)
Domini Social Equity (P)	Pax World Value (P)
Dreyfus Premier Third Century (R)	Pax World Women's Equity (R)
Epiphany Faith and Family Values 100 (P)	Sentinel Sustainable Core Opportunities (P)
Flex-Funds Total Return Utilities (R)	TIAA CREF Inst Social Choice Equity (P)
Green Century Equity (P)	Vanguard FTSE Social Index Fund (R)
Integrity Growth and Income (R)	Vanguard FTSE Social Index Fund (R)
Legg Mason Partners Social Awareness (P)	Walden Social Balanced (P)
LKCM Acquisas Growth (R)	Walden Social Equity (P)

(a) Table 9: Longevity analysis.

(b) Table 10: List of employment funds.

Persistence of abnormal returns and the SRI demand channel.

Read. Table 9 shows that abnormal returns persist for several years but fade by the fifth year. Table 10 lists SRI funds with labor or employment screens.

Economic message. The return pattern is consistent with gradual learning about an intangible asset. The SRI fund list sets up the test of whether fund demand mechanically explains the alpha.

10.6 Table 11: Holdings by employment funds and Table 12: Valuation regressions

Table 11

Holdings by employment funds.

Regressions of a stock's aggregate ownership by employment funds at year-end on a dummy variable for whether the firm was in the most recent list of the "100 Best Companies to Work For in America" (*BC*) and various control variables. *SIZE* is log market equity, *MB* is the log market-to-book ratio, *PRINV* is the inverse of the stock price, *NASDAQ* and *SP500* are dummy variables for inclusion in the Nasdaq and S&P 500 indexes (all measured at year-end), *STD* is the standard deviation of daily stock returns, and *MORET* is the average monthly return (all measured over the year). In specifications (3) and (4), I include *YEARBC*, defined as $(Year - 1984) \times BC$. The coefficients are estimated using a panel regression with industry and year dummies. Robust standard errors are clustered at the industry level. *t*-Statistics are in parentheses. The sample period is 1984–2009. Columns 1 and 3 consider funds that use positive or restrictive employment screens. Columns 2 and 4 only consider funds that employ positive employment screens alone. All coefficients are multiplied by 1,000.

	(1)	(2)	(3)	(4)
	All funds	Positive funds	All funds	Positive funds
<i>BC</i>	0.330 (3.27)***	0.328 (3.28)***	−0.101 (−0.62)	−0.083 (−0.52)
<i>YEARBC</i>			0.039 (3.49)***	0.038 (3.35)***
<i>SIZE</i>	0.054 (8.01)***	0.052 (7.75)***	0.053 (7.98)***	0.051 (7.72)***
<i>MB</i>	−0.029 (−2.84)***	−0.030 (−2.84)***	−0.030 (−2.89)***	−0.030 (−2.89)***
<i>PRINV</i>	0.013 (6.58)***	0.012 (6.29)***	0.012 (6.44)***	0.012 (6.15)***
<i>STD</i>	−0.004 (−1.81)*	−0.004 (−1.88)*	−0.004 (−1.82)*	−0.004 (−1.89)*
<i>MORET</i>	−0.002 (−2.96)***	−0.002 (−2.92)***	−0.002 (−2.90)***	−0.002 (−2.85)***
<i>NASDAQ</i>	0.029 (1.32)	0.030 (1.36)	0.029 (1.31)	0.029 (1.34)
<i>SP500</i>	0.001 (0.03)	−0.005 (−0.17)	0.005 (0.17)	−0.001 (−0.03)
Constant	−0.282 (−5.91)***	−0.273 (−5.76)***	−0.284 (−5.97)***	−0.275 (−5.81)***
# obs	143,487	143,487	143,487	143,487

*, Significant at the 10% level; ***, Significant at the 1% level.

This is substantially smaller than the 3.5% annual alpha, and so in-sample purchases cannot explain the excess returns.²⁷

Hypothesis C is that satisfaction has no effect on shareholder value, but the market believes that it has a positive effect (leading to a negative view that it causes

(a) Table 11: Holdings by employment funds.

Table 12

Valuation regressions.

Regressions of a stock's valuation on a dummy variable for whether the firm is in the current list of the "100 Best Companies to Work For in America" (*BC*) and various control variables. The three valuation measures are *MB*, the log market-to-book ratio, *PE*, the log price-to-earnings ratio, and *AVEBITDA*, the log ratio of aggregate value to EBITDA, and measured at the end of each month in which a Best Companies list was published, i.e., March 1984, February 1993, and January 1998–2007. I drop observations for which the denominator of the valuation ratio is negative. The control variables are all measured at December of the previous year: *ROE* is the return on equity, *FROE*, *F2ROE*, and *F3ROE* are the returns on equity for the next three years, *RDSALES* is the ratio of R&D to sales, *RDMISS* is a dummy variable for whether R&D is missing, and *SP500* is a dummy variable for inclusion in the S&P 500 index. The coefficients are estimated using Fama-MacBeth (1973). *t*-Statistics are in parentheses.

	(1)	(2)	(3)
	<i>MB</i>	<i>PE</i>	<i>AVEBITDA</i>
<i>BC</i>	0.384 (9.24)***	0.092 (2.14)*	0.145 (3.57)***
<i>ROE</i>	0.006 (0.49)	−0.297 (−1.17)	−0.063 (−1.07)
<i>RDSALES</i>	0.030 (1.09)	2.086 (4.73)***	0.803 (1.81)*
<i>RDMISS</i>	−0.100 (−4.46)***	−0.017 (−0.89)	−0.055 (−2.06)*
<i>SP500</i>	0.583 (6.62)***	0.330 (4.65)***	0.168 (3.64)***
<i>FROE</i>	0.038 (1.25)	0.038 (0.88)	0.006 (0.79)
<i>F2ROE</i>	0.008	0.070	0.016

(b) Table 12: Valuation regressions.

Tests of demand pressure and valuation explanations.

Read. Table 11 shows that employment-screened SRI funds overweight the *Best Companies* firms, but the magnitude is too small to explain the returns. Table 12 shows that *Best Companies* firms trade at valuation premiums, not discounts.

Economic message. The alpha is unlikely to be driven mainly by SRI fund demand or by correction of an initial market penalty for employee-friendly policies.

11 How to use this paper in research

11.1 Useful framing

A useful way to cite this paper is as evidence that intangible organizational capital can be economically important but hard for markets to price. The paper is especially useful for projects on human capital, employee welfare, corporate culture, SRI, market efficiency, and managerial myopia.

11.2 Possible limitations to mention

- The design is not a randomized experiment, so causal interpretation should be cautious.
- The *Best Companies* list captures multiple dimensions of firm quality; satisfaction may be correlated with management quality or culture.
- The sample covers the right tail of satisfaction, so the results may not generalize linearly across all firms.
- The abnormal returns may decline if investors learn to price employee satisfaction more efficiently.

12 Conclusion

Edmans (2011) provides strong evidence that employee satisfaction is associated with superior long-run stock returns. The paper is important because it links human capital theories of the firm with asset pricing evidence. The central lesson is not just that satisfied employees matter, but that the market may struggle to value intangibles even when they are publicly observable, independently verified, and attached to large firms. This makes the paper a key reference for research on intangible capital, stakeholder governance, SRI screens, and market underreaction.